System and Devices (SYS 3)

Tutorial Lecture 2 : Pipeline Hazards

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True/False

Which of the following statements is/are TRUE?

1. RAW data hazard could be reduced by operand forwarding.
2. A normal in-order 5 stage MIPS pipeline can achieve an IPC larger than 1.
3. For a MIPS instruction STR R2, 16(R3), some contents stored in its ID/EX pipeline register will bypass the EX unit directly to EX/MEM pipeline register.
4. A normal 5 stage in order RISC pipeline without operand forwarding can have RAW and WAR hazards.

(A) 1 only (B) 1 and 3 only (C) 2 and 4 only (D) 3 and 4 only

Pipeline Hazards

Given a non-pipelined architecture running at 1.5 GHz, that takes 5 cycles to finish an instruction. You want to make it pipelined with 5 stages. Due to hardware overhead the pipelined design will operate only at 1 GHz. 5% of memory instructions cause a stall of 50 cycles, 30% of branch instruction cause a stall of 2 cycles and load-ALU combinations cause a stall of 1 cycle. Assume that in a given program, there exist 20% of branch instructions and 30% of memory instructions. 10% of instructions are load-ALU combinations. What is the speedup of pipelined design over the non-pipelined design?

Pipeline Hazards

A program has 2000 instructions in the sequence L.D, ADD.D, L.D, ADD.D, ……L.D, ADD.D. The ADD.D instruction depends on the L.D instruction right before it. The L.D instruction depends on the ADD.D instruction right before it. If the program is executed on the 5-stage pipeline, what would be the actual CPI with and without operand forwarding technique?